

# Privacy Amplification by Composite Subsampling

## 1 Introduction

Composition and subsampling [Balle et al., 2018] are the key primitives by which Differential Privacy (DP) [Dwork, 2006] can be adopted for private learning algorithms. Previous work [Balle et al., 2018] focus on the case where the data is tabular and unstructured (i.e. sets of elements), or in the case where we observe a single subsampling step [Schuchardt et al., 2024]. In the structured data setting, more involved subsampling schemes involving multiple logically distinct steps are leveraged to construct inputs to private learning algorithms [Huang et al., 2025]. In this work, we consider the privacy properties of composite subsampling mechanisms.

### 1.1 Contributions

Our contributions are as follows. We...

1. Develop a general, unified technique for proving privacy bounds for composite subsampling mechanisms.
2. Use this framework to analyze several concrete problem instances, in particular private training of graph neural networks.

### 1.2 Related Work

We will work on this section later.

## 2 Preliminaries

**Notation.** We follow notation in parallel with [Schuchardt et al., 2024]. Fix a constant  $N$ . Consider a sequence of measurable spaces  $((\mathcal{X}_k, \mathcal{F}_k))_{0 \leq k \leq N+1}$ , each with a corresponding probability space  $(\Omega_k, \mathcal{F}_k, \mathbb{P}_k)$ . Each space  $\mathcal{X}_0, \dots, \mathcal{X}_N$  corresponds to an instance of a generalized batch space (which will generally vary, depending on the problem instance), while the last space  $\mathcal{X}_{N+1}$  corresponds to the output space. To emphasize this fact, we generally set  $\mathcal{Z} = \mathcal{X}_{N+1}$ .

*Example.* Consider the setting of a large database  $\mathcal{D}$  indexed by  $n$ , and consider an index set  $K = [m_1] \times [m_2] \times \dots \times [m_n]$ , such that each entry in  $\mathcal{D}$  can be indexed by  $\mathcal{D}_{\mathbf{k}}$  with  $\mathbf{k} \in K$ . In this setting, each  $\mathcal{X}_k$  corresponds to the space of sets of entries sharing the first  $k$  indices.

*Example.* Following the sampling scheme proposed in Chua et al. [2025], we can let  $\mathcal{X}_1$  denote the space of datasets, let  $\mathcal{X}_2$  denote the space of batches indexed by timestep  $t$ , and allow the base mechanism to be the process of learning from the  $t^{\text{th}}$  batch in the  $t^{\text{th}}$  iteration.

We consider individual randomized subsampling schemes of the form  $S^k : \mathcal{X}_{k-1} \times \Omega_{k-1} \rightarrow \mathcal{X}_k$  and a randomized base mechanism  $B : \mathcal{X}_n \times \Omega_N \rightarrow \mathcal{Z}$ . Naturally, we then define a randomized mechanism as the composition of each subsampling scheme and the base mechanism.

**Definition 2.1** (Composite Subsampling Mechanism). Given a sequence of subsampling mechanisms  $(S_k)_{0 \leq k \leq N}$  and base mechanism  $B$ , a *composite mechanism*  $M : \mathcal{X}_0 \times \Omega \rightarrow \mathbb{Z}$  is defined as

$$M = B \circ (S_n \circ \dots \circ S_1), \quad (1)$$

where  $\Omega$  denotes the outcome space of the product probability measure  $(\Omega, \mathcal{F}, \mathbb{P})$  computed from each  $(\Omega_k, \mathcal{F}_k, \mathbb{P}_k)$ . We generally write  $M_x$  to denote the composite mechanism given some fixed dataset  $x \in \mathcal{X}_0$ .

We'd like to think about each of these randomized functions in terms of their measures. For each randomized mechanism  $S^k$ , we define a corresponding transition kernel  $P^k : \mathcal{X}_{k-1} \times \mathcal{F}_k \rightarrow [0, 1]$  as

$$P^k(x, S) = \mathbb{P}_k(\omega : S^k(x, \omega) \in S).$$

for measurable  $S \subseteq \mathcal{X}_k$ . For simplicity, we let  $P^{N+1}$  denote the transition kernel associated with the randomized base mechanism  $B$ . We consider a desirable property of these kernels below.

**Definition 2.2** (Markov Chains and the Markov Property). Consider a sequence of random variables  $(X_k)_{0 \leq k \leq t}$  taking values in some measurable space  $(\mathcal{X}, \mathcal{F})$  on some probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ . We say that the sequence of random variables  $(X_k)_{0 \leq k \leq t}$  is a *Markov chain* if the chain satisfies the *Markov property* given as

$$\mathbb{P}(X_t \in S \mid X_{1:t-1}) = \mathbb{P}(X_t \in S \mid X_{t-1}), \quad (2)$$

for any measurable  $S \subseteq \mathcal{X}$ .

Using the probability structure above, we can show that the composition of the kernels (as we've defined them) is a Markov chain on some measurable space. In particular, we have the following proposition.

**Proposition 2.3** (Composite Subsampling as Markov Chain). *Assume all variables are as defined above. Consider the measurable input space  $(\mathcal{X}, \mathcal{F}_{\mathcal{X}})$  defined by*

$$\mathcal{X} = \bigoplus_{k=0}^{N+1} \mathcal{X}_k \quad \mathcal{F}_{\mathcal{X}} = \bigoplus_{k=0}^{N+1} \mathcal{F}_k,$$

where  $\bigoplus_i X_i$  denotes the disjoint union of each  $X_i$ , such that each element is distinguished by the original  $X_i$  from which it belongs. Then  $(X_k)_{0 \leq k \leq N+1}$  is a Markov chain with transition kernel  $Q$  satisfying the restriction  $Q|_{\mathcal{X}_{k-1}} = P^k$ , with all other transition probabilities set to zero.

*Proof.* This proposition is directly verifiable from our construction of the transition kernels  $P^k$  and the Markov property (2).  $\square$

It is valid, then, to analyze composite subsampling as defined in (1) as a Markov chain. We are now ready to analyze the outputs of the full Markov chain, which leads to the following proposition.

**Proposition 2.4.** *Assume the same setting as (2.3). Then the distribution (i.e. composite transition kernel) corresponding to the randomized composite mechanism  $M_x$  is given by  $\mathcal{M} : \mathcal{X}_0 \times \mathcal{F}_{\mathcal{X}} \rightarrow [0, 1]$  and can be expressed as*

$$\mathcal{M}(x_0, S) = \mathbb{P}_{N+1}(\omega : M(x_0, \omega) \in S) = \int_{\mathcal{X}_1} \dots \int_{\mathcal{X}_N} P^1(x_0, dx_1) \dots P^{N+1}(x_N, S).$$

We generally write  $\mathcal{M}_x(S) = \mathcal{M}(x, S)$  to denote the probability distribution.

*Proof.* We refer readers to [Levin et al., 2006] for a proof, which relies on the fact that the composition of transition kernels is also a transition kernel.  $\square$

## 2.1 Differential Privacy

The goal of our work is to prove privacy bounds for the composite mechanism  $M_x$ . Consequently, we briefly review the notion of differential privacy.

**Differential Privacy (DP).** We first recall the notion of  $(\epsilon, \delta)$ -DP.

**Definition 2.5** (Approximate Differential Privacy). Consider a randomized mechanism  $M : \mathcal{X} \times \Omega \rightarrow \mathcal{Z}$ . We say that  $M$  is  $(\epsilon, \delta)$ -DP if for any datasets  $x, x'$  satisfying the adjacency relation  $x \simeq x'$  and any measurable subset  $S \subseteq \mathcal{Z}$ , the corresponding distribution  $\mathcal{M}_x$  satisfies

$$\mathcal{M}_x(S) \leq \mathcal{M}_{x'}(S) e^\epsilon + \delta.$$

We generally work with an equivalent definition using the  $\alpha$ -hockey-stick divergence [Balle et al., 2018], which we define below.

**Definition 2.6** ( $\alpha$ -Hockey-Stick Divergence). Consider two distributions  $P, Q$ . We define the  $\alpha$ -hockey-stick divergence to be

$$H_\alpha(P\|Q) = \sup_{S \in \mathcal{F}} (P(S) - \alpha Q(S)) = \int_{\mathcal{Z}} \left( \frac{dP}{dQ} - \alpha \right)_+ dQ(s),$$

where  $\frac{dP}{dQ}$  denotes the Radon-Nikodym derivative w.r.t  $P, Q$ , assuming that both  $P, Q$  are absolutely continuous with respect to some common measure, like the Lebesgue or counting measure. In the case where the densities for  $P, Q$  exist,  $\frac{dP}{dQ}$  reduces to the ratio of the densities.

It is a simple exercise to show the following proposition.

**Proposition 2.7** ( $\alpha$ -Hockey-Stick Differential Privacy). Assume all the same conditions as (2.5). A randomized mechanism  $M$  is  $(\epsilon, \delta)$ -DP if

$$H_{e^\epsilon}(\mathcal{M}_x\|\mathcal{M}_{x'}) \leq \delta.$$

One nice property of the  $\alpha$ -hockey-stick divergence is its joint convexity, which we consider below.

**Proposition 2.8** (Joint Convexity of  $\alpha$ -Hockey-Stick Divergence). Suppose that we take  $P_1, P_2, Q_1, Q_2$  to be distributions with some weight  $\lambda \in [0, 1]$ . Then for any order  $\alpha \geq 0$ , we have

$$H_\alpha(P_1\lambda + P_2(1 - \lambda)\|Q_1\lambda + Q_2(1 - \lambda)) \leq \lambda H_\alpha(P_1\|Q_1) + (1 - \lambda)H_\alpha(P_2\|Q_2).$$

*Proof.* We refer readers to [Balle et al., 2018] for a proof. □

**Adjacency Relations.** We assign an adjacency relation  $\simeq_{\mathcal{X}_k}$  for each dataset/batch space  $\mathcal{X}_k$ . These relations are generally distinct, since the types of elements in each  $\mathcal{X}_k$  are generally distinct.

If we use the  $H_\alpha$  notion of privacy, our goal is not only to upper bound this divergence, but to do so in terms of *dominating pairs*, which dominate (ideally tightly) the full privacy profile of the mechanism. We introduce this notion below.

**Definition 2.9** (Dominating Pairs [Zhu et al., 2022]). We say that the distributions  $(P, Q)$  are a *dominating pair* for the mechanism  $\mathcal{M}_x$  if, for any  $\alpha > 0$  and  $x \simeq x'$ , we have

$$H_\alpha(P, Q) \geq H_\alpha(\mathcal{M}_x, \mathcal{M}_{x'}).$$

In particular, if there exists some  $x \simeq x'$  such that  $H_\alpha(P, Q) = H_\alpha(\mathcal{M}_x, \mathcal{M}_{x'})$ , then we say that  $(P, Q)$  are a *tightly dominating pair*.

## 2.2 Optimal Transport

We briefly review the notion of the coupling between multiple probability distributions.

**Definition 2.10** (Couplings). A coupling between probability measures  $\mathbb{P}_1, \dots, \mathbb{P}_n$  on a space  $(\mathcal{X}, \mathcal{F})$  is a probability measure  $\Gamma$  on the product space  $(\mathcal{X}, \mathcal{F})^n$  such that  $\Gamma$  the  $n^{\text{th}}$  marginal of  $\Gamma$  is  $\mathbb{P}_k$  for each  $k$ , or  $\Gamma \circ \pi_k^{-1} = \mathbb{P}_k$  for each  $k$ , where  $\pi_k$  is the projection map onto the  $k^{\text{th}}$  coordinate.

## 2.3 Distance Compatibility

Distance compatibility will allow us to minimize the transport cost incurred via [Theorem 2](#). We refer readers to [\[Schuchardt et al., 2024\]](#) for a more detailed exposition of distance compatibility. First, we recall the notion of a distance-compatible coupling on a base adjacency relation  $\simeq_{\mathcal{X}_k}$ .

**Definition 2.11** (Distance-Compatible Coupling). Consider a coupling  $\Gamma$  between probability measures  $\mu_1, \dots, \mu_n$  on a space  $(\mathcal{X}, \mathcal{F}_{\mathcal{X}})$ . Fix an adjacency relation  $\simeq$  and its corresponding distance measure  $d(\cdot, \cdot)$ . We say that  $\Gamma$  is distance-compatible with respect to  $\simeq$  and a reference index  $i$ , if for any  $\mathbf{x} \in \text{supp}(\Gamma)$ , we have both of the following.

1. For any index  $u \neq i$ ,  $d(x_i, x_u) = d(x_i, \text{supp}(\mu_u))$ .
2. For any indices  $u, v \neq i$ ,  $d(x_u, x_v) = d(\text{supp}(\mu_u), \text{supp}(\mu_v))$ .

Intuitively, distance compatibility ensures that  $\Gamma$  only assigns mass to elements that are as close to each other and to the reference index as possible.

## 2.4 Privacy Amplification by Subsampling

As described in [Section 1](#), the notion of *privacy amplification by subsampling* observes that differentially private algorithms have stronger privacy guarantees when run on a subset of a fixed dataset, versus the complete dataset. We realize the (randomized) generation of these subsets via the successive application of subsampling.

Composite subsampling schemes are of particular interest to learning scenarios involving structured data (e.g. graphs), and learning algorithms have multiple steps of subsampling, such as subgraph construction, edge sampling, etc.

In particular, reusing the notation from [Section 2](#), we are interested in considering the composition of a differentially private base mechanism and a composite subsampling scheme  $M_x = B \circ (S_n \circ \dots \circ S_1)$ . Letting  $\mathcal{M}_x$  denote the output probability distribution of  $M_x$ , we'd like to bound the quantity  $H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'})$ .

# 3 Constructing upper bounds for divergences between composite mechanisms

## 3.1 General Tools

Assume the same notation as in [Section 2](#) and the same goal of bounding  $H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'})$  as in [Section 2.4](#). Since we have established in [Proposition 2.3](#) that we can view the randomized mechanism  $M$  as a Markov chain, we can apply some tools from Markov chain theory to aid our analysis.

First, we recall that in our construction of a Markov chain in [Proposition 2.3](#), we generated a transition kernel  $Q$  satisfying the restriction property  $Q|_{\mathcal{X}_{k-1}} = P^k$ . Based on our analysis of  $\mathcal{M}_x$ , it is not hard to see that we can express the distribution as the successive application of  $Q$  on a delta distribution, that is,  $\mathcal{M}_x = \delta_x Q^N$ . We can then consider the two following facts.

**Lemma 1.** *Given an initial measure  $\mu_0 : \mathcal{F}_{\mathcal{X}_0} \rightarrow [0, 1]$  and some  $0 \leq t \leq N$ , we have*

$$\mu_0 Q^N = \mu_t Q^{N-t},$$

*with  $\mu_t = \mu_0 Q^t$ . Consequently, running the Markov chain for  $t$  steps starting from  $\mu_0$  is equivalent to running the Markov chain for  $N - t$  steps starting from  $\mu_t$ . Then, we can relate upper bounds on  $H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'})$  to upper bounds on partially-run Markov chains, as stated in the following theorem.*

In our setting  $\mu_0 = \delta_x$ , so we have  $\mu_t = \delta_x Q^t$  and can write  $\mathcal{M}_x = \mu_t Q^{N-t}$  for any  $t \in \{1, \dots, N\}$ .

**Theorem 1.** For a randomized mechanism  $M_x$  with distribution  $\mathcal{M}_x$  of a Markov chain with transition kernel  $Q$ , we have

$$H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'}) = H_{e^\epsilon}(\mu_t Q^{N-t} \| \mu'_t Q^{N-t}),$$

with  $\mu_t = \delta_x Q^t$  and  $\mu'_t = \delta_{x'} Q^t$ .

We can now proceed using a conditioning strategy to generate a class of upper bounds on  $H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'})$ , as analogues to the approach exploited in [Schuchardt et al., 2024].

## 3.2 Conditional Upper Bounds

In this section, we discuss two separate strategies for bounding the divergences between distributions arising from composite subsampling. The first involves a more challenging coupling construction with simpler inner terms, while the second involves a simpler coupling construction with more challenging inner terms.

### 3.2.1 Coupling Conditional Distributions

**Theorem 2.** Assume the same setting as Proposition 2.3. Consider a randomized subsampling mechanism  $M_x = B \circ (S_n \circ \dots \circ S_1)$  with base mechanism  $B$  and subsampling mechanisms  $(S_k)_{0 \leq k \leq N}$ . Fix some time  $t \in \{1, \dots, N\}$ , and consider two disjoint partitions of the event space until time  $t$ , such that

$$\bigcup_{i=1}^I A_i = \bigoplus_{k=1}^t \mathcal{X}_k \quad \text{and} \quad \bigcup_{j=1}^J B_j = \bigoplus_{k=1}^t \mathcal{X}_k.$$

Let  $\Gamma$  be a coupling between the conditional distributions  $\mu_t(\cdot | A_1), \dots, \mu_t(\cdot | A_I)$  and  $\mu'_t(\cdot | B_1), \dots, \mu'_t(\cdot | B_J)$ , for a total of  $I + J$  probability measures. Then we have

$$H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \int_{\mathcal{X}^{I+J}} H_{e^\epsilon} \left( \sum_{i=1}^I Q^{N-t}(\mathbf{x}_i, \cdot) \mu_t(A_i) \parallel \sum_{j=1}^J Q^{N-t}(\mathbf{x}_{I+j}, \cdot) \mu'_t(B_j) \right) d\Gamma(\mathbf{x}).$$

*Proof.* Fix  $t$ , so that we have  $\mathcal{M}_x = \mu_t Q^{n-t}$ . Then, using the law of total probability, we have

$$\begin{aligned} \mu_t Q^{n-t} &= \sum_{i=1}^I \left( \int_{\mathcal{X}} Q^{n-t}(x, \cdot) d\mu_t(x | A_i) \right) \mu_t(A_i) \\ &= \sum_{i=1}^I \left( \int_{\mathcal{X}} Q^{n-t}(x, \cdot) d(\Gamma \circ \pi_t^{-1})(x) \right) \mu_t(A_i). \end{aligned}$$

Let  $\mathbf{x} = (x_1, \dots, x_I, x_{I+1}, \dots, x_{I+J})$ , such that the first  $I$  coordinates correspond to each of the conditional distributions  $\mu_t(\cdot | A_i)$ , and the last  $J$  coordinates correspond to each of the conditional distributions  $\mu'_t(\cdot | B_j)$ . We can rewrite the inner term above using this projection, along with linearity of integration, as

$$\begin{aligned} \mu_t Q^{n-t} &= \sum_{i=1}^I \left( \int_{\mathcal{X}} Q^{n-t}(x, \cdot) d(\Gamma \circ \pi_t^{-1})(x) \right) \mu_t(A_i) \\ &= \sum_{i=1}^I \left( \int_{\mathcal{X}^{I+J}} Q^{n-t}(\mathbf{x}_i, \cdot) d\Gamma(\mathbf{x}) \right) \mu_t(A_i) \\ &= \int_{\mathcal{X}^{I+J}} \left( \sum_{i=1}^I Q^{n-t}(\mathbf{x}_i, \cdot) \mu_t(A_i) \right) d\Gamma(\mathbf{x}). \end{aligned}$$

Similarly, for  $\mathcal{M}_{x'}$ , we have

$$\mu'_t Q^{n-t} = \int_{\mathcal{X}^{I+J}} \left( \sum_{j=1}^J Q^{n-t}(\mathbf{x}_{I+j}, \cdot) \mu'_t(B_j) \right) d\Gamma(\mathbf{x}).$$

Then  $\mathcal{M}_x = \mu_t Q^{n-t}$  and  $\mathcal{M}_{x'} = \mu'_t Q^{n-t}$  are expectations with respect to the same measure  $\Gamma$ , so joint convexity of the hockey-stick divergence gives

$$H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \int_{\mathcal{X}^{I+J}} H_{e^\epsilon} \left( \sum_{i=1}^I Q^{n-t}(\mathbf{x}_i, \cdot) \mu_t(A_i) \left\| \sum_{j=1}^J Q^{n-t}(\mathbf{x}_{I+j}, \cdot) \mu'_t(B_j) \right\| \right) d\Gamma(\mathbf{x}).$$

what we wanted to show.  $\square$

At first glance, one could view the structure of [Theorem 2](#) as an optimization problem (i.e. find the optimal  $t$  that constructs the tightest upper bound). However, it is more natural to consider  $t$  arising from the original adjacency relation for the first layer of dataset/batch space. In particular, two primary approaches for using [Theorem 2](#) are as follows.

1. Take  $t = N$ . Then,  $Q^{N-t} = Q$  for any  $x \in \mathcal{X}_N$ , but by construction,  $Q|_{\mathcal{X}_n} = P^{N+1}$ , where  $P^{N+1}$  is the transition kernel associated with the randomized base mechanism  $B$ . In this setting, partitioning of the event space only affects the weights of the mixture, and bounding the worst-case divergence usually involves considering worst-case inputs and high-level information about the base mechanism  $B$ , such as its  $\ell_2$  sensitivity.
2. Take  $t < N$ . Then,  $Q^{N-t}$  represents several layers of the transition kernel applied after time  $t$ . In this regime, the partitions  $A_i$  and  $B_j$  can capture coarser or more structured information about the intermediate layers of the process, and the resulting bound can incorporate finer details of the mechanism's action on partially subsampled datasets.

Let  $c_t(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$  denote the inner divergence term in [Theorem 2](#). Then, we can rewrite the upper bound as

$$H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \int_{\mathcal{X}^{I+J}} c_t(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) d\Gamma(\mathbf{x}).$$

It remains to bound  $c_t(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$ . Naturally, following the approach in [[Schuchardt et al., 2024](#)], we can consider the worst-case. In particular, we consider

$$\hat{c}_t(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) = \sup_{\hat{\mathbf{x}}^{(1)}, \hat{\mathbf{x}}^{(2)}} c_t(\hat{\mathbf{x}}^{(1)}, \hat{\mathbf{x}}^{(2)}),$$

where the maximization is taken over all  $\hat{\mathbf{x}} = (\hat{\mathbf{x}}^{(1)}, \hat{\mathbf{x}}^{(2)})$  satisfying, for any  $k, l \in \{1, 2\}$ ,

$$d(\hat{\mathbf{x}}^{(k)}, \hat{\mathbf{x}}^{(l)}) \leq d(\mathbf{x}^{(k)}, \mathbf{x}^{(l)}),$$

Clearly  $\hat{c}_t(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) \geq c_t(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$ , so we have

$$H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \int_{\mathcal{X}^{I+J}} \hat{c}_t(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) d\Gamma(\mathbf{x}).$$

Of course, this still leaves open the construction of a coupling  $\Gamma$  that minimizes the right hand side. We show that a distance-compatible coupling appropriately captures the right notion of minimality, and therefore minimizes the right hand side of this upper bound. In particular, we have the following proposition.

**Proposition 3.1.** *Consider the setting of [Theorem 2](#). Fix a reference index  $i \in \{1, \dots, I + J\}$ . For  $u \in \{1, \dots, I + J\}$ , define the  $u^{\text{th}}$  conditional law*

$$s_u = \begin{cases} \mu_t(\cdot | A_u) & \text{if } 1 \leq u \leq I, \\ \mu'_t(\cdot | B_{u-I}) & \text{if } I + 1 \leq u \leq I + J. \end{cases}$$

Let  $\Gamma$  be a coupling of  $(s_u)_{u=1}^{I+J}$  on  $\mathcal{X}_t^{I+J}$ . If  $\Gamma$  is distance-compatible with respect to  $\simeq_{\mathcal{X}_t}$  and reference index  $i$ , then

$$\Gamma \in \arg \min_{\Gamma'} \int_{\mathcal{X}_t^{I+J}} \hat{c}_t(\mathbf{x}) d\Gamma'(\mathbf{x}),$$

where the minimum is over all couplings  $\Gamma'$  of  $(s_u)_{u=1}^{I+J}$ .

*Proof.* We first record a basic inequality relating pointwise distances and distances to supports. For any probability measures  $\nu_1, \nu_2$  on  $(\mathcal{X}_t, \mathcal{F}_t)$  and any  $x, y \in \mathcal{X}_t$ , we have

$$d(x, y) \geq d(x, \text{supp}(\nu_2)) \geq d(\text{supp}(\nu_1), \text{supp}(\nu_2)),$$

where  $d(x, \text{supp}(\nu)) = \inf_{z \in \text{supp}(\nu)} d(x, z)$  and  $d(\text{supp}(\nu_1), \text{supp}(\nu_2)) = \inf_{z_1 \in \text{supp}(\nu_1), z_2 \in \text{supp}(\nu_2)} d(z_1, z_2)$ . See [Schuchardt et al., 2024] for further details.

Now fix an arbitrary coupling  $\Gamma'$  of  $(s_u)_{u=1}^{I+J}$ , and write  $\mathbf{x} = (x_1, \dots, x_{I+J}) \in \mathcal{X}_t^{I+J}$ . By definition of the worst-case cost, we have

$$\hat{c}_t(\mathbf{x}) = \sup_{\hat{\mathbf{x}} \in \mathcal{X}_t^{I+J}} c_t(\hat{\mathbf{x}}) \quad \text{subject to} \quad d(\hat{x}_u, \hat{x}_v) \leq d(x_u, x_v) \text{ for all } u \neq v.$$

Using the inequality above, for any fixed  $\mathbf{x}$  we can relax these constraints to obtain a lower bound. Indeed, the constraints

$$d(\hat{x}_i, \hat{x}_u) \leq d(x_i, \text{supp}(s_u)) \quad \text{for all } u \neq i, \quad d(\hat{x}_u, \hat{x}_v) \leq d(\text{supp}(s_u), \text{supp}(s_v)) \quad \text{for all } u, v \neq i, u \neq v$$

are implied by  $d(\hat{x}_u, \hat{x}_v) \leq d(x_u, x_v)$  in the sense that the right-hand sides above are no larger than  $d(x_u, x_v)$  for every pair, and hence define a feasible region that is at least as large. Consequently,

$$\hat{c}_t(\mathbf{x}) \geq \kappa_t^{(i)}(x_i),$$

where we define the reference-index cost

$$\kappa_t^{(i)}(x_i) = \sup_{\hat{\mathbf{x}} \in \mathcal{X}_t^{I+J}} c_t(\hat{\mathbf{x}})$$

subject to

$$d(\hat{x}_i, \hat{x}_u) \leq d(x_i, \text{supp}(s_u)) \text{ for all } u \neq i, \quad d(\hat{x}_u, \hat{x}_v) \leq d(\text{supp}(s_u), \text{supp}(s_v)) \text{ for all } u, v \neq i, u \neq v.$$

In particular,  $\kappa_t^{(i)}$  depends only on the reference coordinate  $x_i$ . Integrating the pointwise lower bound with respect to  $\Gamma'$  gives

$$\int_{\mathcal{X}_t^{I+J}} \hat{c}_t(\mathbf{x}) d\Gamma'(\mathbf{x}) \geq \int_{\mathcal{X}_t^{I+J}} \kappa_t^{(i)}(x_i) d\Gamma'(\mathbf{x}) = \int_{\mathcal{X}_t} \kappa_t^{(i)}(x) ds_i(x),$$

where the final equality holds because the  $i$ th marginal of  $\Gamma'$  is  $s_i$  by definition of a coupling.

We now show that any distance-compatible coupling  $\Gamma$  attains this lower bound, which implies optimality. If  $\Gamma$  is distance-compatible with reference index  $i$ , then for every  $\mathbf{x} \in \text{supp}(\Gamma)$  we have

$$d(x_i, x_u) = d(x_i, \text{supp}(s_u)) \text{ for all } u \neq i, \quad d(x_u, x_v) = d(\text{supp}(s_u), \text{supp}(s_v)) \text{ for all } u, v \neq i, u \neq v.$$

For such  $\mathbf{x}$ , the relaxed constraints defining  $\kappa_t^{(i)}(x_i)$  coincide with the original constraints defining  $\hat{c}_t(\mathbf{x})$ , and therefore the relaxation is tight:

$$\hat{c}_t(\mathbf{x}) = \kappa_t^{(i)}(x_i) \quad \text{for all } \mathbf{x} \in \text{supp}(\Gamma).$$

It follows that

$$\int_{\mathcal{X}_t^{I+J}} \hat{c}_t(\mathbf{x}) d\Gamma(\mathbf{x}) = \int_{\mathcal{X}_t^{I+J}} \kappa_t^{(i)}(x_i) d\Gamma(\mathbf{x}) = \int_{\mathcal{X}_t} \kappa_t^{(i)}(x) ds_i(x).$$

Since the right-hand side is a lower bound for every coupling  $\Gamma'$ , the coupling  $\Gamma$  achieves the minimum, as claimed.  $\square$

**Corollary 3.2.** *In the setting of [Theorem 3.1](#), assume there exists a distance-compatible coupling with reference index  $i$ . Then*

$$\min_{\Gamma'} \int_{\mathcal{X}_t^{I+J}} \hat{c}_t(\mathbf{x}) d\Gamma'(\mathbf{x}) = \int_{\mathcal{X}_t} \kappa_t^{(i)}(x) ds_i(x),$$

where  $\kappa_t^{(i)}$  is defined by

$$\kappa_t^{(i)}(x) = \sup_{\hat{\mathbf{x}} \in \mathcal{X}_t^{I+J}} c_t(\hat{\mathbf{x}})$$

subject to

$$d(\hat{x}_i, \hat{x}_u) \leq d(x, \text{supp}(s_u)) \text{ for all } u \neq i, \quad d(\hat{x}_u, \hat{x}_v) \leq d(\text{supp}(s_u), \text{supp}(s_v)) \text{ for all } u, v \neq i, u \neq v.$$

*Proof.* The proof follows immediately from [Theorem 3.1](#). The lower bound

$$\int \hat{c}_t d\Gamma' \geq \int \kappa_t^{(i)} ds_i$$

holds for every coupling  $\Gamma'$ , while any distance-compatible coupling attains equality.  $\square$

In particular, for the balls-in-bins/random allocation model, we can upper bound the coupling weights  $\kappa_t^{(i)}$  via the reduction proposed in [Schuchardt et al. \[2024\]](#) and the dominating pairs used in [Chua et al. \[2025\]](#).

## 4 Constructing Dominating Pairs

We now show how to convert the upper bounds from [Theorem 2](#) into dominating pairs for the full composite mechanism. In particular, each choice of coupling and conditioning scheme induces an upper bound on the hockey-stick divergence at every order  $\alpha > 0$ . This upper bound is therefore a privacy profile. By the standard realization result for privacy profiles, there exists a pair of distributions achieving exactly this profile, which then forms a dominating pair for the mechanism.

### 4.1 Realizing the privacy profile

For a randomized mechanism  $M$  and adjacent inputs  $x \simeq x'$ , we consider the notion of the *privacy profile* of the mechanism.

**Definition 4.1** (Privacy profile). A function  $\delta : (0, \infty) \rightarrow [0, 1]$  is called a privacy profile if there exist distributions  $P, Q$  such that for any  $\alpha > 0$ , we have

$$\delta(\alpha) = H_\alpha(P\|Q).$$

We use the following standard fact.

**Proposition 4.2** (Realization of privacy profiles). *Let  $\delta : (0, \infty) \rightarrow [0, 1]$  be a privacy profile. Then there exist distributions  $P, Q$  on a measurable space  $(\mathcal{Y}, \mathcal{G})$  such that*

$$H_\alpha(P\|Q) = \delta(\alpha) \quad \text{for all } \alpha > 0.$$

Consequently, to construct a dominating pair for a mechanism, it suffices to produce a pointwise upper bound on its privacy profile.



## 4.2 Dominating pairs from conditional couplings

We first treat the main theorem from [Theorem 2](#).

**Theorem 3.** *Assume the setting of [Theorem 2](#). Fix  $t \in \{1, \dots, N\}$ , disjoint partitions  $(A_i)_{i=1}^I$  and  $(B_j)_{j=1}^J$ , and a coupling  $\Gamma$  between the conditional laws*

$$\mu_t(\cdot \mid A_1), \dots, \mu_t(\cdot \mid A_I), \quad \mu'_t(\cdot \mid B_1), \dots, \mu'_t(\cdot \mid B_J).$$

For each  $\mathbf{x} = (x_1, \dots, x_{I+J}) \in \mathcal{X}_t^{I+J}$ , define the two distributions

$$R_{\mathbf{x}} = \sum_{i=1}^I \mu_t(A_i) Q^{N-t}(x_i, \cdot), \quad R'_{\mathbf{x}} = \sum_{j=1}^J \mu'_t(B_j) Q^{N-t}(x_{I+j}, \cdot).$$

Now, define

$$\delta_{\Gamma}(\alpha) = \int_{\mathcal{X}_t^{I+J}} H_{\alpha}(R_{\mathbf{x}} \| R'_{\mathbf{x}}) d\Gamma(\mathbf{x}), \quad \alpha > 0.$$

Then  $\delta_{\Gamma}$  dominates the privacy profile of the composite mechanism; that is, for any  $\alpha > 0$ , we have  $H_{\alpha}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \delta_{\Gamma}(\alpha)$ . Consequently, any pair  $(P_{\Gamma}, Q_{\Gamma})$  satisfying  $H_{\alpha}(P_{\Gamma} \| Q_{\Gamma}) = \delta_{\Gamma}(\alpha)$  is a dominating pair for the mechanism  $M$ .

*Proof.* The bound

$$H_{\alpha}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \int_{\mathcal{X}_t^{I+J}} H_{\alpha}(R_{\mathbf{x}} \| R'_{\mathbf{x}}) d\Gamma(\mathbf{x})$$

is exactly the conclusion of [Theorem 2](#), with  $e^{\epsilon}$  replaced by a general order  $\alpha > 0$ . Thus, for all  $\alpha > 0$ , we have

$$H_{\alpha}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \delta_{\Gamma}(\alpha)$$

It follows that  $\delta_{\Gamma}$  is an upper bound on the privacy profile of the mechanism. By [Theorem 4.2](#), there exist distributions  $P_{\Gamma}, Q_{\Gamma}$  such that for all  $\alpha > 0$ , we have

$$H_{\alpha}(P_{\Gamma} \| Q_{\Gamma}) = \delta_{\Gamma}(\alpha)$$

Therefore

$$H_{\alpha}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq H_{\alpha}(P_{\Gamma} \| Q_{\Gamma}),$$

from which it follows that  $(P_{\Gamma}, Q_{\Gamma})$  is a dominating pair for the mechanism  $\mathcal{M}_x$ .  $\square$

This theorem, although relatively trivial, helps us understand how to construct a dominating pair for the composite mechanism.

## 4.3 Dominating pairs from pointwise inner dominating pairs

The previous theorem is abstract because it leaves the inner divergence  $H_{\alpha}(R_{\mathbf{x}} \| R'_{\mathbf{x}})$  unevaluated. In practice, we will often have an explicit dominating pair for this inner mixture mechanism. The next corollary records this reduction.

**Corollary 4.3.** *Assume the setting of [Theorem 3](#). Suppose that for each  $\mathbf{x} \in \mathcal{X}_t^{I+J}$  there exists a pair of distributions  $(P_{\mathbf{x}}, Q_{\mathbf{x}})$  such that for all  $\alpha > 0$ , we have*

$$H_{\alpha}(R_{\mathbf{x}} \| R'_{\mathbf{x}}) \leq H_{\alpha}(P_{\mathbf{x}} \| Q_{\mathbf{x}})$$

Letting  $\bar{\delta}(\alpha) = \int_{\mathcal{X}_t^{I+J}} H_{\alpha}(P_{\mathbf{x}} \| Q_{\mathbf{x}}) d\Gamma(\mathbf{x})$ , we have for all  $\alpha > 0$ ,

$$H_{\alpha}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \bar{\delta}(\alpha).$$

Hence any pair  $(\bar{P}, \bar{Q})$  realizing  $\bar{\delta}$  is a dominating pair for the composite mechanism  $\mathcal{M}_x$ .

*Proof.* By assumption,

$$H_\alpha(R_{\mathbf{x}}\|R'_{\mathbf{x}}) \leq H_\alpha(P_{\mathbf{x}}\|Q_{\mathbf{x}}) \quad \text{for all } \mathbf{x}, \alpha.$$

Integrating with respect to  $\Gamma$  and applying [Theorem 3](#) yields

$$H_\alpha(\mathcal{M}_x\|\mathcal{M}_{x'}) \leq \int H_\alpha(P_{\mathbf{x}}\|Q_{\mathbf{x}}) d\Gamma(\mathbf{x}) = \bar{\delta}(\alpha).$$

The final claim again follows from [Theorem 4.2](#).  $\square$

Therefore, when given a dominating pair for the inner mixture mechanism, we can construct a dominating pair for the composite mechanism. Our next task will be to show that these dominating pairs can satisfy some notion of optimality via distance compatibility.

#### 4.4 Distance-compatible couplings

We now combine the previous construction with [Theorem 3.1](#). Recall that

$$c_t(\mathbf{x}) = H_\alpha(R_{\mathbf{x}}\|R'_{\mathbf{x}})$$

and that  $\hat{c}_t(\mathbf{x})$  denotes its worst-case relaxation under the induced distance constraints. When  $\Gamma$  is distance-compatible, [?? 3.1?? 3.2](#) show that the resulting averaged upper bound is minimal among all couplings with the same marginals. This immediately yields an optimal dominating-profile construction within this relaxation class.

**Theorem 4.** *Assume the setting of [Theorem 3.2](#), and fix  $\alpha > 0$ . Suppose  $\Gamma$  is distance-compatible with reference index  $i$ . Define*

$$\delta_i(\alpha) = \int_{\mathcal{X}_t} \kappa_{t,\alpha}^{(i)}(x) ds_i(x),$$

where

$$\kappa_{t,\alpha}^{(i)}(x) = \sup_{\hat{\mathbf{x}} \in \mathcal{X}_t^{I+J}} H_\alpha \left( \sum_{u=1}^I \mu_t(A_u) Q^{N-t}(\hat{x}_u, \cdot) \parallel \sum_{v=1}^J \mu'_t(B_v) Q^{N-t}(\hat{x}_{I+v}, \cdot) \right)$$

subject to

$$d(\hat{x}_i, \hat{x}_u) \leq d(x, \text{supp}(s_u)) \text{ for all } u \neq i,$$

and

$$d(\hat{x}_u, \hat{x}_v) \leq d(\text{supp}(s_u), \text{supp}(s_v)) \text{ for all } u, v \neq i, u \neq v.$$

Then

$$H_\alpha(\mathcal{M}_x\|\mathcal{M}_{x'}) \leq \delta_i(\alpha) \quad \text{for all } \alpha > 0.$$

Moreover, among all couplings of the conditional laws  $(s_u)_{u=1}^{I+J}$ , the profile  $\delta_i$  is minimal within the relaxation defined by  $\hat{c}_t$ . Hence any pair realizing  $\delta_i$  is a dominating pair for the full mechanism.

*Proof.* By construction,  $\hat{c}_t$  upper bounds the original inner divergence pointwise. Therefore,

$$H_\alpha(\mathcal{M}_x\|\mathcal{M}_{x'}) \leq \int \hat{c}_t(\mathbf{x}) d\Gamma'(\mathbf{x})$$

for any coupling  $\Gamma'$  of the conditional laws. By [?? 3.1?? 3.2](#), if  $\Gamma$  is distance-compatible with reference index  $i$ , then the minimum of the right-hand side over all such couplings equals

$$\int_{\mathcal{X}_t} \kappa_{t,\alpha}^{(i)}(x) ds_i(x) = \delta_i(\alpha).$$

Thus, for all  $\alpha > 0$ , we have  $H_\alpha(\mathcal{M}_x\|\mathcal{M}_{x'}) \leq \delta_i(\alpha)$ . The existence of a realizing dominating pair follows from [Theorem 4.2](#).  $\square$

## 4.5 A discrete mixture form

In many applications, the quantity  $\kappa_{t,\alpha}^{(i)}(x)$  depends on  $x$  only through a finite statistic, such as the number of affected records, edges, or sampled blocks that survive to the base mechanism. In that case, we show that the dominating profile reduces to a finite mixture.

**Corollary 4.4.** *Assume the setting of [Theorem 4](#). Suppose there exists a measurable statistic  $T : \mathcal{X}_t \rightarrow \{0, 1, \dots, m\}$  and pairs of distributions  $(P_r, Q_r)_{r=0}^m$  such that*

$$\kappa_{t,\alpha}^{(i)}(x) = H_\alpha(P_{T(x)} \| Q_{T(x)}) \quad \text{for all } x \in \mathcal{X}_t, \alpha > 0.$$

*Let  $w_r = s_i(T = r)$ ,  $r = 0, \dots, m$ . Then the privacy profile of the full composite mechanism is bounded by*

$$H_\alpha(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \sum_{r=0}^m w_r H_\alpha(P_r \| Q_r).$$

*Hence any pair realizing the profile  $\delta(\alpha) = \sum_{r=0}^m w_r H_\alpha(P_r \| Q_r)$  is a dominating pair for the full mechanism.*

*Proof.* This is immediate from [Theorem 4](#) after decomposing the integral defining  $\delta_i(\alpha)$  according to the level sets of  $T$ .  $\square$

This corollary will likely let us connect to existing subsampling analyses. Once the inner mechanism has known dominating pairs indexed by an exposure statistic, the outer composite subsampling scheme contributes only the weights  $(w_r)_{r=0}^m$ .

## 5 Solving the Derived Optimization Problem

## 6 Constructing Mechanism-Specific Group Privacy Bounds

### 6.1 Reduction to Existing Privacy Bounds

We show that [Theorem 2](#) is a strict generalization of [Theorem 3.4](#) in [\[Schuchardt et al., 2024\]](#).

**Theorem 5.** *Let  $M = B \circ S$  be a randomized subsampling mechanism with base mechanism  $B$  and subsampling mechanism  $S$ . Letting  $t = 1$  recovers [Theorem 3.4](#), and consequently all relevant results in [\[Schuchardt et al., 2024\]](#).*

*Proof.* Letting  $t = 1$  in [Theorem 2](#) for  $M = B \circ S$  gives us the upper bound

$$H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \int_{\mathcal{X}^{I+J}} H_{e^\epsilon} \left( \sum_{i=1}^I Q(\mathbf{x}_i, \cdot) \mu_1(A_i) \left\| \sum_{j=1}^J Q(\mathbf{x}_{I+j}, \cdot) \mu'_1(B_j) \right\| \right) d\Gamma(\mathbf{x}).$$

Note that  $\mu_1 = P_{S_x}(A_i)$  and  $\mu'_1 = P_{S_{x'}}(B_j)$ , or the transition kernels corresponding to the subsampling mechanisms  $S_x$  and  $S_{x'}$ . Since the final application of the transition kernel  $Q$  is just the base mechanism  $B$ , we can write  $Q(x, \cdot) = b_x(\cdot)$ . In this form, we have

$$H_{e^\epsilon}(\mathcal{M}_x \| \mathcal{M}_{x'}) \leq \int_{\mathcal{X}^{I+J}} H_{e^\epsilon} \left( \sum_{i=1}^I b_{\mathbf{x}_i} \cdot P_{S_x}(A_i) \left\| \sum_{j=1}^J b_{\mathbf{x}_{I+j}} \cdot P_{S_{x'}}(B_j) \right\| \right) d\Gamma(\mathbf{x}),$$

which is precisely [Theorem 3.4](#) in [\[Schuchardt et al., 2024\]](#). All other results follow through the same ansatz outlined in their paper.  $\square$

## 6.2 Simple Examples in Database Schemes

Now, we consider a simple example in the database setting to make use of our framework and exposition regarding [Theorem 2](#).

Consider the setting of a large database  $\mathcal{D}$  indexed by  $n$ , and consider an index set  $K = [m_1] \times [m_2] \times \dots \times [m_n]$ , such that each entry in  $\mathcal{D}$  can be indexed by  $\mathcal{D}_{\mathbf{k}}$  with  $\mathbf{k} \in K$ . When referring to a set of entries sharing the first  $t$  indices, we refer to this group as  $\mathcal{D}_{k_1, \dots, k_t, \cdot} = \mathcal{D}_{\mathbf{k}_t}$  for notational simplicity.

Consider a composite subsampling scheme to sample and learn from individual entries from this database  $M = B \circ (S_n \circ \dots \circ S_1)$ . In particular, let each  $S_t$  perform Poisson subsampling with rate  $r_t$  for the next index  $k_{t+1}$  from the subset of  $\mathcal{D}_{\mathbf{k}_t}$  that have been selected in the previous sampling steps.

Our approach will be to define events *at the last level which they are concerned*. Referring to the example above, if we wish to preserve the privacy of families, our event structure in  $A_i$  will only deal with the first layer of subsampling steps. Consequently, it is natural to set  $t = 1$  in [Theorem 2](#) and analyze accordingly. If we wish to preserve the privacy of individuals, then our event structure will deal with the second layer of subsampling steps (which inherently rely on the first), so we set  $t = 2$ . If we wish to preserve the privacy of specific entries, then our event structure will need to deal with each step of the subsampling procedure, and consequently select  $t = 3$ .

**Theorem 6.** *Assume the setting above. Consider  $x \simeq x'$  at the  $p^{\text{th}}$  level  $\mathcal{D}_{\mathbf{k}_p}$ , such that  $x$  differs from  $x'$  from some entry at the  $p^{\text{th}}$  index onwards, where  $p$  is some integer. Then  $H_\alpha(\mathcal{M}_x \| \mathcal{M}_{x'})$  is upper bounded by*

$$\max_{\mathbf{x}} H_\alpha \left( Q^{N-p}(\mathbf{x}_0, \cdot) (1 - q_p) + Q^{N-p}(\mathbf{x}_1, \cdot) \cdot q_p \parallel Q^{N-p}(\mathbf{x}'_0, \cdot) (1 - q_p) + Q^{N-p}(\mathbf{x}'_1, \cdot) \cdot q_p \right),$$

where  $q_p = \prod_{i=1}^{p-1} r_i$ .

*Proof.* Let  $A_0$  denote the event that the differing entry is not sampled in the  $p^{\text{th}}$  layer, and  $A_1$  denote the event that the differing entry is sampled in the  $p^{\text{th}}$  layer. Analogously define  $B_0$  and  $B_1$ .

It is not hard to see that

$$\mu(A_0) = 1 - \prod_{i=1}^{p-1} r_i, \quad \mu(A_1) = \prod_{i=1}^{p-1} r_i,$$

since each ancestor entry must be sampled at rate  $r_i$  for the  $i^{\text{th}}$  step for the differing entry to be sampled in the  $p^{\text{th}}$  layer. Similarly, we have

$$\mu(B_0) = 1 - \prod_{i=1}^{p-1} r_i, \quad \mu(B_1) = \prod_{i=1}^{p-1} r_i.$$

Let  $\mathbf{x}^{(1)} = (\mathbf{x}_0, \mathbf{x}_1)$  and  $\mathbf{x}^{(2)} = (\mathbf{x}'_0, \mathbf{x}'_1)$ . Now, define the coupling  $\Gamma$  between  $\mu(\cdot|A_0), \mu(\cdot|A_1), \mu(\cdot|B_0), \mu(\cdot|B_1)$  as

$$\Gamma(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) = \begin{cases} \mu(\mathbf{x}_0|A_0) & \text{if } \mathbf{x}_0 = \mathbf{x}'_0, \mathbf{x}_1 \simeq \mathbf{x}'_1 \\ 0 & \text{otherwise.} \end{cases}$$

We first show that this is a valid coupling by computing marginals. First, consider the projection onto the first coordinate  $\pi((\mathbf{x}^{(1)}, \mathbf{x}^{(2)})) = \mathbf{x}_0$ . Then, we have

$$\sum_{(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) \in \mathcal{X}_p^4} \chi \left( \pi((\mathbf{x}^{(1)}, \mathbf{x}^{(2)})) = \mathbf{x}_0 \right) \Gamma(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) = \mu(\mathbf{x}_0|A_0),$$

since there is only one possible value for  $\mathbf{x}_0$  that satisfies the projection and has  $\mu(\mathbf{x}_0|A_0) > 0$ . Similarly, we have

$$\sum_{(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) \in \mathcal{X}_p^4} \chi \left( \pi((\mathbf{x}^{(1)}, \mathbf{x}^{(2)})) = \mathbf{x}_1 \right) \Gamma(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) = \mu(\mathbf{x}_1|A_1),$$

by taking the summation over all possible indices. Computations for the marginals for  $\mu(\cdot|B_0)$  and  $\mu(\cdot|B_1)$  are analogous. It follows that  $\Gamma$  is a valid coupling. Distance compatibility follows from the proof in [Schuchardt et al., 2024].

Consequently, using the worst-case divergence bound with  $\mu(A_0) = \mu(B_0) = 1 - \prod_{i=1}^{p-1} r_i$  and  $\mu(A_1) = \mu(B_1) = \prod_{i=1}^{p-1} r_i$ , with Theorem 2, the result follows.  $\square$

With our framework, it is substantially more challenging to work in the group privacy setting, as computing the probabilities  $\mu(A_i)$  and  $\mu(B_j)$  relies on a complicated summation (that can be derived, but is not trivial to compute), since the probabilities of sampling a particular entry depend heavily on where individual entries are located in the database.

## 7 Learning from Graphs

Below, we outline the basic process of learning iteratively from graphs. The goal of this analysis is to provide perspective on the “right” way to learn from sampled subgraphs.

We assume, for right now, that we are attempting to do *graph representation learning*, with a focus on *node-level* tasks. That is, we want to learn an embedding for each node in a graph.

Suppose that we have a graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ . For some fixed integer  $d$ , let  $\mathbf{X} \in \mathbb{R}^{d \times |\mathcal{V}|}$  denote the features of each node. If we let  $\mathbf{h}_u^{(k)}$  represent the hidden embedding at each node  $u \in \mathcal{V}$  at the  $k^{\text{th}}$  iteration, we aggregate information as

$$\mathbf{h}_u^{(k+1)} = \text{UPDATE}^{(k)} \left( \mathbf{h}_u^{(k)}, \text{AGGREGATE}^{(k)}(\{\mathbf{h}_v^{(k)}, \forall v \in \mathcal{N}(u)\}) \right)$$

where  $\mathcal{N}(u)$  represents the neighborhood of  $u$  in  $\mathcal{G}$ . After running this for iterations  $\{1, \dots, K\}$ , we consider the embeddings  $\mathbf{z}_u = \mathbf{h}_u^{(K)}$ . Suppose we are given an appropriate loss function  $\ell : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$  and  $m$  subgraphs  $\{S_{v_i}\}_{i=1}^m$  rooted at  $v_i \in \mathcal{V}$ . Then, we update the weights parameterizing the update and aggregation steps,  $\Theta_t$  during each minibatch as

$$\Theta_{t+1} \leftarrow \Theta_t - \eta \left( \frac{1}{m} \sum_{i=1}^m \nabla \ell(\mathbf{y}_{v_i}, \mathbf{z}_{v_i}(S_{v_i})) \right),$$

where  $\eta$  denotes the (arbitrary) learning rate. Crucially, the losses are only computed at the central nodes of the sampled subgraphs, not *all* of the nodes in the sampled subgraph.

## 8 Subgraph Generation

We briefly overview the process of subgraph generation. Let  $G = (V, E)$  be a graph, directed or undirected. For each  $v \in V$ , let  $N(v) \subseteq V$  denote the neighbor set used by the sampling procedure, meaning out-neighbors, in-neighbors, or symmetrized neighbors depending on the chosen convention. A minibatch subsampling procedure is a randomized map that takes  $G$  and returns a smaller computation object on which a forward and backward pass is executed, typically anchored by a seed set  $S$  specifying the nodes or edges on which the loss is evaluated. The objective of subgraph generation is to produce a node set  $U \supseteq S$  and a set of sampled edges  $F \subseteq E \cap (U \times U)$  so that learning can proceed using only  $(U, F)$ .

A generic decomposition of essentially all minibatch subsampling schemes can be expressed by separating the selection of seeds, the iterative expansion rule, and the assembly rule that materializes the returned computation object. Fix a depth parameter  $L \in \mathbb{N}$  and per-layer budgets  $(m_\ell)_{\ell=1}^L$ . Let  $\mathcal{S}$  be a distribution over seed sets  $S \subseteq V$  or seed edge sets  $S \subseteq E$ , with the understanding that edge seeds can be converted to a node seed set by taking endpoints. For each layer  $\ell$ , let  $\text{Sel}_\ell$  be a local selection kernel that, given  $(v, m_\ell, G)$ ,

returns a finite set or multiset  $T_\ell(v) \subseteq N(v)$ . The sampler proceeds by drawing  $S \sim \mathcal{S}$ , setting the layer- $L$  frontier  $F_L := S$ , and for  $\ell = L, L-1, \dots, 1$  expanding each  $v \in F_\ell$  via  $T_\ell(v)$  and updating the frontier by

$$F_{\ell-1} := F_\ell \cup \bigcup_{v \in F_\ell} T_\ell(v).$$

The final sampled node set is  $U := F_0$ . The sampled edge set is obtained through an assembly rule **Assemble**, which may return a layered computation graph  $(U, (E_\ell)_{\ell=1}^L)$  with

$$E_\ell := \{(u, v) \in E : v \in F_\ell, u \in T_\ell(v)\},$$

or may return the induced subgraph  $G[U]$ . We outline some forms of sampling below.

**Uniform Sampling.** Fix a node  $v$  with degree  $d_v = |N(v)|$  and a budget  $m$ . Uniform sampling without replacement is specified by taking  $\text{Sel}_\ell(v, m, G)$  to return a subset  $T(v) \subseteq N(v)$  of size  $\min\{m, d_v\}$  chosen uniformly from all such subsets. When  $d_v \geq m$ , this means  $\Pr[T(v) = T] = \binom{d_v}{m}^{-1}$  for every  $T \subseteq N(v)$  with  $|T| = m$ . Uniform sampling with replacement is specified by drawing i.i.d. samples  $U_1, \dots, U_m$  from the uniform distribution on  $N(v)$ , meaning  $\Pr[U_i = u] = 1/d_v$  for  $u \in N(v)$ , and returning  $T(v)$  as the resulting multiset, possibly deduplicated depending on the implementation.

**Weighted Sampling.** Weighted neighbor sampling is specified by choosing nonnegative weights  $w_{vu} \geq 0$  for  $u \in N(v)$  and defining a distribution  $q_v$  on  $N(v)$  by  $q_v(u) = w_{vu} / \sum_{u' \in N(v)} w_{vu'}$ . Weighted sampling with replacement takes  $\text{Sel}_\ell(v, m, G)$  to draw i.i.d. samples from  $q_v$  and return the resulting multiset. Weighted sampling without replacement takes  $\text{Sel}_\ell(v, m, G)$  to return a size- $m$  subset with inclusion preference increasing in  $w_{vu}$ . In either case, the decomposition is unchanged, with only the local kernel  $\text{Sel}_\ell$  differing.

**Random-Walk Sampling.** Random-walk-based subsampling is obtained by defining  $\text{Sel}_\ell(v, m, G)$  through short random walks starting from  $v$ . A typical kernel runs one or more length- $r$  walks, optionally with restart, collects visited nodes, and returns the first  $m$  distinct visited nodes as  $T_\ell(v)$ . The resulting sampler still follows the same frontier expansion and assembly steps, with the selection kernel now reflecting nodes and edges visited along the walk.

**Induced Subgraph Sampling.** Node-induced and edge-induced subgraph samplers shift randomness away from iterative frontier expansion and into the direct selection of a node set  $U$  or an edge set whose incident nodes define  $U$ . In the generic decomposition, this corresponds to choosing  $\mathcal{S}$  to output  $S = U$  directly, choosing  $L = 0$  so that no expansion occurs, and choosing **Assemble** to return the induced subgraph  $G[U]$ . Node-induced samplers specify  $\mathcal{S}$  as a distribution over node subsets, while edge-induced samplers specify  $\mathcal{S}$  as a distribution over edge subsets and then define  $U$  as the set of endpoints.

**Cluster Sampling.** Cluster-based minibatching is obtained by first computing a partition  $V = \bigcup_i C_i$  into disjoint clusters. In the generic decomposition, we can choose  $\mathcal{S}$  to sample an index  $k$  and output  $S = C_k$  or a small union of clusters, sets  $L = 0$ , and returns  $G[S]$  via the induced-subgraph assembly rule. Variants that include boundary information can be expressed by modifying **Assemble** to add selected cut edges or by taking  $S$  to include neighboring clusters, while keeping the same decomposition.

## 9 Balls and bins approach for Graph Structured Data

Graph neural networks face an inherent problem in training under differential privacy that is absent from standard DP-SGD on independent data. This is that the message-passing mechanism of graphs entangles gradient contributions across various nodes. Protecting a single node, means affecting the gradients of the nodes surrounding it.

With standard DP-SGD, every loss term  $\nabla \ell_v$  is clipped to the norm  $C$ . Because node  $u$  affects up to  $|\mathcal{R}_{\text{out}}(u)|$  clipped terms, where  $|\mathcal{R}_{\text{out}}(u)|$  is the number of nodes in the outgoing receptive field, the  $\ell_2$  sensitivity of the clipped gradient sum with respect to node  $u$  is bounded by  $|\mathcal{R}_{\text{out}}(u)| \cdot C$ . In a graph with max degree of  $K_{\text{max}}$ , this gives us a worst case sensitivity of  $K_{\text{max}} \cdot C$ . This means for very large or dense GNNs, we would require significant accuracy degradations to achieve meaningful privacy.

## 9.1 Degree-Independent Node-Level DP for GNN Training via Directed Subgraph Partitioning

### 9.1.1 Directed Subgraph Partitioning

We provide a construction for degree-independent sensitivity by combining three ideas: (i) AGNNCERT’s node-centric directed graph division, (ii) user-level clipping, and (iii) coverage-corrected gradient aggregation. Following AGNNCERT’s node-centric division, every undirected edge  $(u, v) \in E$  becomes two directed edges  $u \rightarrow v$  and  $v \rightarrow u$ . Through directed message passing, node  $v$  updates using only its incoming messages:  $h_v^{(\ell)} = \text{Combine}\left(h_v^{(\ell-1)}, \text{Aggregate}\left(\left\{h_u^{(\ell-1)} : u \rightarrow v \in \vec{E}\right\}\right)\right)$ . Each node and its corresponding outgoing edges is then assigned randomly to one of  $N$  isolated bins. Therefore, the gradients in one bin, do not affect another’s.

### 9.1.2 The Mechanism

Let  $T$  denote the total number of training iterations and  $N$  denote the number of bins. For each training iteration  $t = 1, \dots, T$ :

1. (Optional) Epoch Assignment. At the start of an epoch, randomly subsample which iteration a node gets assigned to. Each iteration has roughly  $|G|/T$  nodes, and the following steps will be applied to only nodes within that iteration.
2. Bin Assignment ( $S_1^{(t)}$ ). Each node  $v \in V$  independently selects a bin index.  $b^{(t)}(v) \sim \text{Uniform}\{1, \dots, N\}$  independently for all  $v \in V$ . This repeats at every iteration  $t$ .
3. Directed Subgraph Construction. For each bin  $i \in 1, \dots, N$ , construct the directed subgraph representation  $\vec{G}_i^{(t)}$ .  $\vec{G}_i^{(t)} = (V, \vec{E}_i^{(t)}, X)$ , where  $\vec{E}_i^{(t)} = \{u \rightarrow v : u \in B_i^{(t)}, (u, v) \in E\}$ , and  $B_i^{(t)} = \{v \in V : b^{(t)}(v) = i\}$  is the set of nodes assigned to bin  $i$  at iteration  $t$ . All nodes in  $V$  are present in every subgraph, but only nodes in  $B_i^{(t)}$  have outgoing edges in the subgraph,  $\vec{G}_i^{(t)}$ .
4. Forward Pass. For each bin, run the  $k$ -layer GNN forward pass on  $\vec{G}_i^{(t)}$ , using directed message passing. Each node  $v$  computes its representation  $h_{v,i}^{(k)}$  by aggregating incoming messages within the nodes in its bin.
5. Loss and Gradient Computation. Compute loss for each node  $v \in B_i^{(t)}$  assigned to that bin.  $g_i^{(t)} = \sum_{v \in B_i^{(t)}} \nabla_{\theta} \ell(h_{v,i}^{(k)}, y_v)$ .
6. Coverage Correction. Apply a coverage correction factor to compensate for the restricted neighborhood  $\tilde{g}_i^{(t)} = c_i^{(t)} \cdot g_i^{(t)}$ , where  $c_i^{(t)} = \frac{1}{|B_i^{(t)}|} \sum_{v \in B_i^{(t)}} \frac{d_v^{(i)}}{d_v}$ , where  $d_v$  is the degree of  $v$  in the full graph  $G$  and  $d_v^{(i)}$  is the in-degree of  $v$  in the subgraph, as in the number of  $v$ ’s neighbors assigned to bin  $i$ . This correction factor adjusts the sampling bias introduced by the partition. It is applied before the noise and clipping of DP-SGD, allowing it to not affect the sensitivity.
7. Per-Bin Clipping. Clip each bin’s corrected gradient to norm  $C$ :  $\hat{g}_i^{(t)} = \text{clip}(\tilde{g}_i^{(t)}, C) = \tilde{g}_i^{(t)} \cdot \min\left(1, \frac{C}{\|\tilde{g}_i^{(t)}\|_2}\right)$ .
8. Aggregation and Noise. Average the clipped gradients across bins and add Gaussian noise:  $\bar{g}^{(t)} = \frac{1}{N} \sum_{i=1}^N \hat{g}_i^{(t)} + \mathcal{N}\left(0, \sigma^2 \cdot \frac{4C^2}{N^2} \cdot I\right)$ .
9. Model Update. Update the parameters of the model  $\theta^{(t+1)} = \theta^{(t)} - \eta \cdot \bar{g}^{(t)}$ .

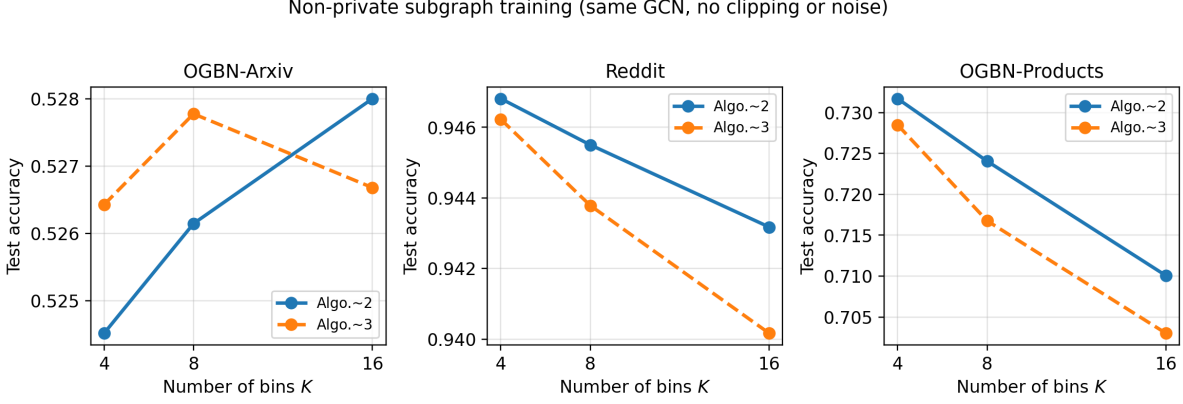


Figure 1: Non-private subgraph training: test accuracy vs.  $K$  on the three benchmarks. Solid = Algo. 2; dashed = Algo. 3.

### 9.1.3 Analysis

For two node-adjacent graphs  $G, G'$  differing by a single node  $u$ . At iteration  $t$ , node  $u$  is assigned to exactly one bin say  $b^{(t)}(u) = i$ . In every other subgraph, node  $u$  has no outgoing edges. Therefore, by the directed message passing mechanism,  $u$ 's features do not propagate to any other node's representations. Consequently, the bin gradients  $g_j^{(t)}$  and their clipped outputs are identical under  $G$  and  $G'$  for all  $j \neq i$ . In bin  $i$ , removing  $u$  changes the forward pass, removes  $u$ 's loss term, and thus changes  $g_i^{(t)}$ .

However, because we clip the bin aggregate to norm  $C$ , we have  $\|\hat{g}_i^{(t)}(G)\|_2 \leq C$  and  $\|\hat{g}_i^{(t)}(G')\|_2 \leq C$ , so by the triangle inequality  $\|\hat{g}_i^{(t)}(G) - \hat{g}_i^{(t)}(G')\|_2 \leq \|\hat{g}_i^{(t)}(G)\|_2 + \|\hat{g}_i^{(t)}(G')\|_2 \leq 2C$ .

Since we then get the final gradient by averaging over  $N$  bins and only one bin is affected:  $\|\bar{g}^{(t)}(G) - \bar{g}^{(t)}(G')\|_2 = \frac{1}{N} \|\hat{g}_i^{(t)}(G) - \hat{g}_i^{(t)}(G')\|_2 \leq \frac{2C}{N}$ . This bound holds for any graph topology, any node degree, and any GNN depth  $k$ . It is further lowered by the privacy amplifications of the random bin assignment.

## 10 Experiments

We study transductive node classification with the same two-layer subgraph GCN everywhere. Hyperparameters include: number of bins  $K \in \{4, 8, 16\}$ , single-phase training-node sampling rate  $q \in \{0.05, 0.10\}$ , gradient clip  $C=1$ , Gaussian noise multiplier  $\sigma/C$ , and PRV accounting at  $\delta=10^{-5}$  (reported  $\epsilon$ ). Full-graph GCN/MLP baselines are omitted from this draft; where helpful we compare against the *non-DP subgraph* runs (Algo. 2 and 3 without noise) so partitioning and subsampling effects are isolated from privacy noise.

### 10.1 Utility without differential privacy

We first measure *partitioning cost* when no DP mechanism is applied: same optimizer and early-stopping protocol, but no per-step clipping or noise. Figure 1 shows test accuracy versus  $K$  for Algo. 2 (internal edges per bin) and Algo. 3 (same with dummy-bin dropout  $p=0.3$ ). Table 1 lists the same numbers (means over seeds when multiple logs exist).



Table 1: Non-DP subgraph test accuracy (mean over seeds). Only  $K \in \{4, 8, 16\}$ .

| Dataset       | $K$ | Algo. 2 | Algo. 3 |
|---------------|-----|---------|---------|
| ogbn-arxiv    | 4   | 0.525   | 0.526   |
| ogbn-arxiv    | 8   | 0.526   | 0.528   |
| ogbn-arxiv    | 16  | 0.528   | 0.527   |
| reddit        | 4   | 0.947   | 0.946   |
| reddit        | 8   | 0.946   | 0.944   |
| reddit        | 16  | 0.943   | 0.940   |
| ogbn-products | 4   | 0.732   | 0.728   |
| ogbn-products | 8   | 0.724   | 0.717   |
| ogbn-products | 16  | 0.710   | 0.703   |

## 10.2 Differential privacy and sampling rate $q$

Under DP we use single-phase Poisson subsampling of training nodes with per-step inclusion probability  $q$ . To avoid overcrowded curves, the *main* DP figures fix  $K=8$  and show only four trajectories: Algo. 2 vs. Algo. 3, each at  $q \in \{0.05, 0.10\}$ , with noise multiplier on the horizontal axis (top panel) and PRV  $\varepsilon$  on the horizontal axis (bottom panel). A dotted line marks the best non-DP subgraph accuracy from Section 10.1.

Table 2 summarizes a *scalar* comparison at  $\sigma/C=1$  and  $K=8$ : how much tightening  $q$  from 0.10 to 0.05 buys in  $\varepsilon$  at similar accuracy.

## 10.3 Effect of the number of bins $K$ under DP

Finer partitions change both subgraph structure and the effective sampling rate per step ( $q/K$  in single-phase accounting), so we report a compact table at a fixed privacy-noise operating point:  $\sigma/C=1$  and  $q=0.10$ . Table 3 contrasts  $K \in \{4, 8, 16\}$  (Algos. 2 and 3);  $\varepsilon$  differs across  $K$  because the same nominal noise is applied per bin while composition uses the reported step count.

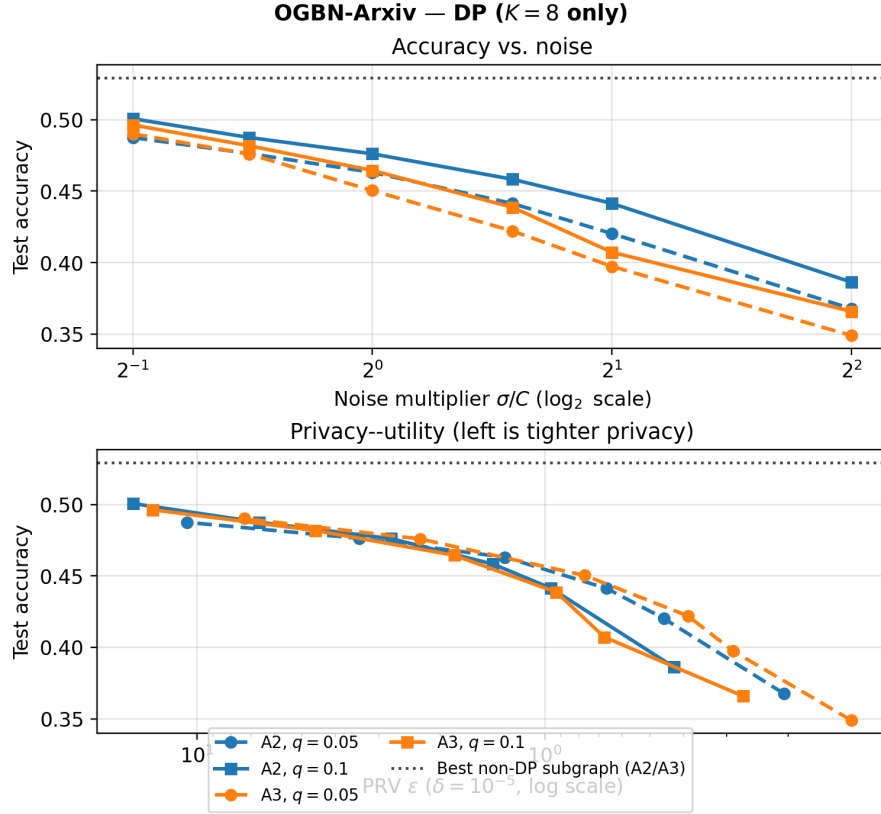


Figure 2: **OGBN-Arxiv**,  $K=8$ . DP test accuracy vs. noise (top) and vs. PRV  $\epsilon$  (bottom).

Table 2: DP at  $\sigma/C=1$ ,  $K=8$ : test accuracy and PRV  $\epsilon$  for both  $q$  values.

| Dataset       | $q$  | Algo. 2 acc. | Algo. 3 acc. | $\epsilon$ (A2) | $\epsilon$ (A3) |
|---------------|------|--------------|--------------|-----------------|-----------------|
| ogbn-arxiv    | 0.05 | 0.463        | 0.450        | 1.302           | 0.768           |
| ogbn-arxiv    | 0.1  | 0.476        | 0.464        | 2.765           | 1.821           |
| reddit        | 0.05 | 0.872        | 0.849        | 0.749           | 0.549           |
| reddit        | 0.1  | 0.884        | 0.877        | 1.499           | 1.088           |
| ogbn-products | 0.05 | 0.656        | 0.656        | 1.018           | 0.669           |

Table 3: DP at  $\sigma/C=1$ ,  $q=0.10$ : test accuracy and  $\epsilon$  vs.  $K$ . Rows omitted if that setting is not yet present in the run logs (e.g. incomplete sweeps).

| Dataset    | $K$ | Algo. 2 acc. | Algo. 3 acc. | $\epsilon$ (A2) | $\epsilon$ (A3) |
|------------|-----|--------------|--------------|-----------------|-----------------|
| ogbn-arxiv | 4   | 0.458        | 0.432        | 6.832           | 3.781           |
| ogbn-arxiv | 8   | 0.476        | 0.464        | 2.765           | 1.821           |
| ogbn-arxiv | 16  | 0.492        | 0.484        | 1.209           | 0.860           |
| reddit     | 4   | 0.893        | 0.879        | 3.832           | 2.651           |
| reddit     | 8   | 0.884        | 0.877        | 1.499           | 1.088           |
| reddit     | 16  | 0.901        | 0.871        | 0.757           | 0.539           |

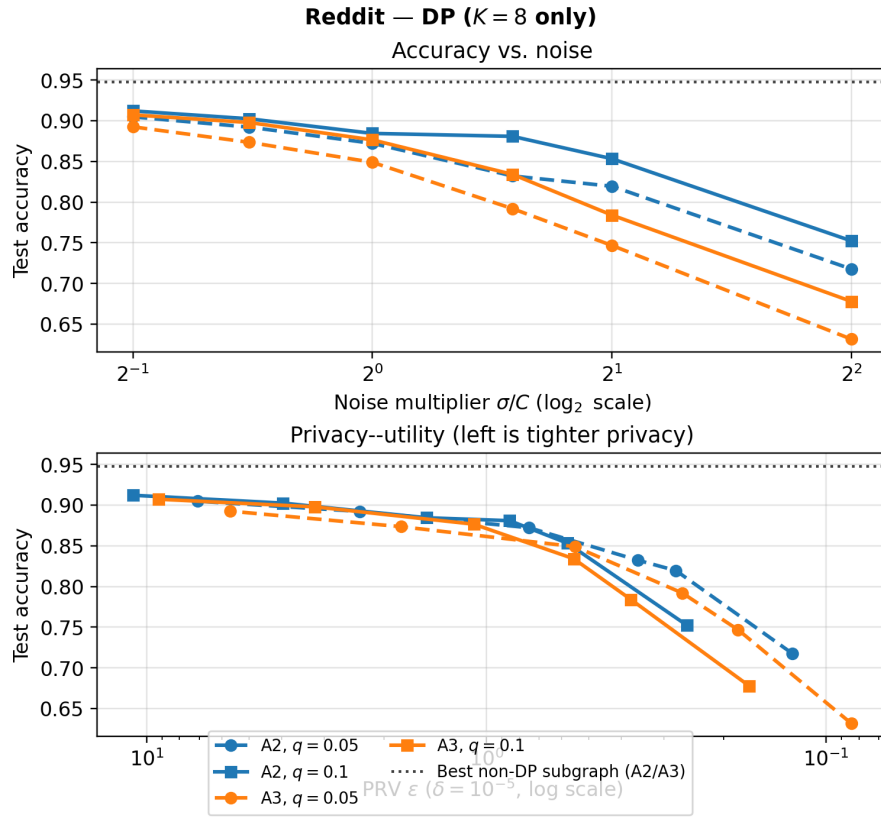


Figure 3: **Reddit**,  $K=8$ . Same layout as Figure 2.

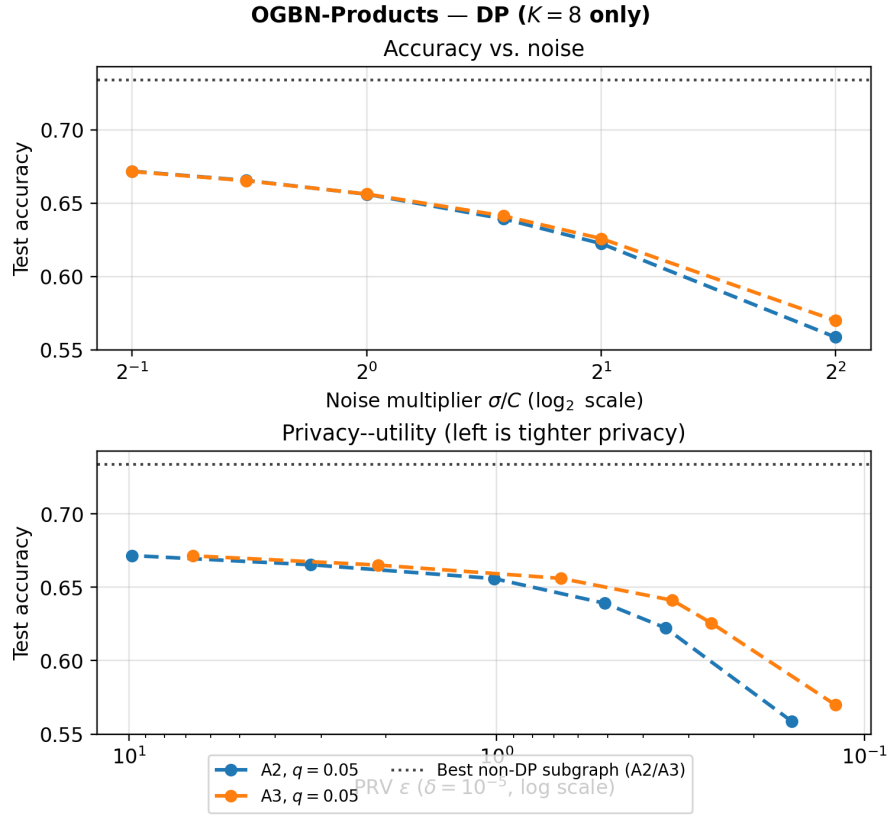


Figure 4: **OGBN-Products**,  $K=8$ . Same layout as Figure 2.

## 11 Subsampling Algorithms

### 11.1 Iteration-level

**AS:** I wasn't able to figure out a way to deal with the multiple-contribution issue of nodes-in-bins.

#### 11.1.1 Basic Nodes-and-Bins

Assign nodes to bins, also add all outgoing neighbors.

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**Algorithm 1** Nodes-and-Bins, Basic

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**Require:** Nodes  $\mathcal{V} = \{1, \dots, N\}$ , edges  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ , number of bins  $K$

```
for  $i \in \mathcal{V}$  do
   $y_i \sim \text{Uniform}(\{1, \dots, K\})$  # Bin assignment
end for
for  $k \in \{1, \dots, K\}$  do
   $\mathcal{V}_k \leftarrow \{\}$ 
   $\mathcal{E}_k \leftarrow \{\}$ 
  for  $i \in \mathcal{V}$  do
    if  $y_i \neq k$  then
      continue
    else
       $\mathcal{V}_k \leftarrow \mathcal{V}_k \cup \{i\}$  # Add node itself
      for  $j : (i, j) \in \mathcal{E}$  do
         $\mathcal{V}_k \leftarrow \mathcal{V}_k \cup \{j\}$  # Add its outgoing neighbors
         $\mathcal{E}_k \leftarrow \mathcal{E}_k \cup \{(i, j)\}$ 
      end for
    end if
  end for
end for
return  $\{(\mathcal{V}_1, \mathcal{E}_1), \dots, (\mathcal{V}_K, \mathcal{E}_K)\}$ 
```

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Later use: Compute per-sample gradients, **aggregate into per-bin gradients**, clip, noise, aggregate.

#### 11.1.2 Removing sinks

We discussed the following: Removing nodes that only receive messages (have ingoing edges) but don't send messages (have outgoing messages).

Since we only add outgoing edges for node  $i$  when it has been directly (and not indirectly) assigned to bin  $k$ , this is equivalent so just constructing the graph restricted to nodes that have been directly assigned to bin  $k$ .

Later use: Compute per-sample gradients, **aggregate into per-bin gradients**, clip, noise, aggregate.

#### 11.1.3 Removing sinks and subsampling

In both of the previous algorithms, there is at least one bin with a modified gradient. For stronger amplification, we can create an additional dummy bin " $\perp$ " that is not used for gradient computation.

Later use: Compute per-sample gradients, **aggregate into per-bin gradients while discarding the dummy bin**, clip, noise, aggregate. wha

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**Algorithm 2** Nodes-and-Bins, Removing sinks

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**Require:** Nodes  $\mathcal{V} = \{1, \dots, N\}$ , edges  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ , number of bins  $K$

```
for  $i \in \mathcal{V}$  do
   $y_i \sim \text{Uniform}(\{1, \dots, K\})$  # Bin assignment
end for
for  $k \in \{1, \dots, K\}$  do
   $\mathcal{V}_k \leftarrow \{\}$ 
   $\mathcal{E}_k \leftarrow \{\}$ 
  for  $i \in \mathcal{V}$  do
    if  $y_i \neq k$  then
      continue
    else
       $\mathcal{V}_k \leftarrow \mathcal{V}_k \cup \{i\}$  # Add node itself
      for  $j : (i, j) \in \mathcal{E}$  do
        if  $y_j \neq k$  then
          continue
        end if
         $\mathcal{V}_k \leftarrow \mathcal{V}_k \cup \{j\}$  # Add its outgoing neighbors
         $\mathcal{E}_k \leftarrow \mathcal{E}_k \cup \{(i, j)\}$ 
      end for
    end if
  end for
end for
return  $\{(\mathcal{V}_1, \mathcal{E}_1), \dots, (\mathcal{V}_K, \mathcal{E}_K)\}$ 
```

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**Algorithm 3** Nodes-and-Bins, Removing sinks, Subsampled

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**Require:** Nodes  $\mathcal{V} = \{1, \dots, N\}$ , edges  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ , number of bins  $K$

```
for  $i \in \mathcal{V}$  do
   $y_i \sim \text{Uniform}(\{1, \dots, K\})$  # Bin assignment
   $u \sim \text{Bernoulli}(p_\perp)$ 
  if  $u = 1$  then
     $y_i \leftarrow \perp$ 
  end if
end for
for  $k \in \{1, \dots, K\}$  do
   $\mathcal{V}_k \leftarrow \{\}$ 
   $\mathcal{E}_k \leftarrow \{\}$ 
  for  $i \in \mathcal{V}$  do
    if  $y_i \neq k$  then
      continue # Note that we always continue for  $\perp$  nodes
    else
       $\mathcal{V}_k \leftarrow \mathcal{V}_k \cup \{i\}$  # Add node itself
      for  $j : (i, j) \in \mathcal{E}$  do
        if  $y_j \neq k$  then
          continue # Note that we always continue for  $\perp$  nodes
        end if
         $\mathcal{V}_k \leftarrow \mathcal{V}_k \cup \{j\}$  # Add its outgoing neighbors
         $\mathcal{E}_k \leftarrow \mathcal{E}_k \cup \{(i, j)\}$ 
      end for
    end if
  end for
end for
return  $\{(\mathcal{V}_1, \mathcal{E}_1), \dots, (\mathcal{V}_K, \mathcal{E}_K)\}$ 
```

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